

Exact Domain-Wall Ground-State Benchmark for a Lattice Chern Insulator

Asad and GPT-5.4 Codex

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This note collects the formulas used in the notebook `notebooks/exact_DW_GS_benchmark.ipynb` and organizes them into a single theoretical narrative. The benchmark concerns an exactly solvable Gaussian ground state of a real-space domain-wall Chern-insulator Hamiltonian, its momentum-resolved and boundary-resolved spectra, the entanglement contour, fixed- x squared-correlation profiles, the local Chern marker, strip-integrated contour diagnostics, and a contour-restricted multipartite-information construction. All figures presently stored in `DW_GS_Benchmark_Figs/` are included with captions.

I. MODEL AND DOMAIN-WALL GEOMETRY

The notebook studies a two-orbital free-fermion model on an $N_x \times N_y$ square lattice. At each unit cell (x, y) one introduces the spinor

$$\Psi_{x,y} = \begin{pmatrix} c_{x,y,0} \\ c_{x,y,1} \end{pmatrix}, \quad (1)$$

and uses the site-orbital flattening convention

$$i(\mu, x, y) = \mu + 2x + 2N_x y, \quad \mu \in \{0, 1\}. \quad (2)$$

The notebook convention is

$$\alpha_{\text{triv}} \equiv \alpha_1, \quad \alpha_{\text{top}} \equiv \alpha_2, \quad (3)$$

with the domain-wall slab generated by the code path `create_domain_wall`. In the default construction the mass profile is

$$\alpha(x, y) = \begin{cases} \alpha_{\text{top}}, & x_0 \leq x \leq x_1, \\ \alpha_{\text{triv}}, & \text{otherwise,} \end{cases} \quad (4)$$

where the two domain-wall positions are stored as

$$\text{DW_loc} = [x_0, x_1]. \quad (5)$$

In the code this slab is centered near $x = N_x/2$ and its width is chosen as an $\mathcal{O}(0.2N_x)$ fraction of the system size.

The real-space Hamiltonian used by the exact benchmark is

$$H = \sum_{x,y} \Psi_{x,y}^\dagger [\alpha(x, y) \sigma_z] \Psi_{x,y} + \sum_{x,y} \left(\Psi_{x,y}^\dagger t_x \Psi_{x+1,y} + \Psi_{x,y}^\dagger t_y \Psi_{x,y+1} + \text{h.c.} \right), \quad (6)$$

with nearest-neighbor matrices

$$t_x = -\frac{1}{2}\sigma_z - \frac{i}{2}\sigma_x, \quad t_y = -\frac{1}{2}\sigma_z - \frac{i}{2}\sigma_y. \quad (7)$$

For a spatially uniform mass α this corresponds to the familiar Bloch Dirac-Chern form

$$h(\mathbf{k}) = \sin k_x \sigma_x + \sin k_y \sigma_y + (\alpha - \cos k_x - \cos k_y) \sigma_z. \quad (8)$$

II. CONTINUUM EDGE EXPECTATION: CHIRAL FREE FERMION IN 1 + 1D

The domain-wall benchmark is motivated by the expectation that each interface supports a low-energy chiral boundary mode. In continuum language, the universal fixed point is a chiral free fermion conformal field theory in 1 + 1 dimensions. The chiral fermion field ψ has scaling dimension

$$h = \frac{1}{2}, \quad (9)$$

so on a spatial circle of circumference L its equal-time two-point function has the standard chord-length form

$$\langle \psi^\dagger(y) \psi(0) \rangle \propto \frac{1}{\sin(\pi y/L)}, \quad (10)$$

up to a phase and a nonuniversal amplitude. Consequently, the squared magnitude obeys

$$|\langle \psi^\dagger(y) \psi(0) \rangle|^2 \propto \sin^{-2}(\pi y/L). \quad (11)$$

This is the continuum reason that a lattice observable dominated by a single chiral edge mode is expected to yield slope -2 when plotted as $\log C$ versus the logarithm of a periodic chord length. The squared correlation diagnostics below test precisely how well that expectation is realized, and over what spatial range in x it remains valid near the two domain walls.

The notebook also exposes an optional flag `triv_region_local_mode`. In that branch the complement of the domain-wall slab is replaced by decoupled onsite modes, with onsite block $-\mathbb{I}_2$, and all hoppings touching that region are removed. Since the figures stored in the benchmark directory were generated from the standard branch, the main text below focuses on Eqs. (4)–(7).

III. EXACT GAUSSIAN GROUND STATE

The equilibrium covariance is obtained by diagonalizing the single-particle Hamiltonian,

$$H u_n = E_n u_n, \quad (12)$$

forming the occupied-state projector

$$P_- = \sum_{E_n < 0} |u_n\rangle \langle u_n|, \quad (13)$$

and finally constructing the flattened covariance matrix

$$G = 2P_- - \mathbb{I}. \quad (14)$$

This is precisely the object returned by `G.CI_domain_wall`. At half filling, Eq. (14) encodes the entire Gaussian state.

The notebook supplements the raw ordered spectrum of H by a momentum-resolved diagnostic along y . One reshapes the flattened matrix into

$$G(x, y, \mu; x', y', \nu), \quad (15)$$

performs a unitary Fourier transform on the bra y index and an inverse transform on the ket y' index,

$$\tilde{G}(x, k_y, \mu; x', k'_y, \nu) = \frac{1}{N_y} \sum_{y, y'} e^{-ik_y y} G(x, y, \mu; x', y', \nu) e^{+ik'_y y'}, \quad (16)$$

and then extracts the contiguous $(2N_x) \times (2N_x)$ blocks associated with fixed k_y in the notebook's flattening convention. The same procedure is applied to H for the displayed energy bands versus k_y/π .

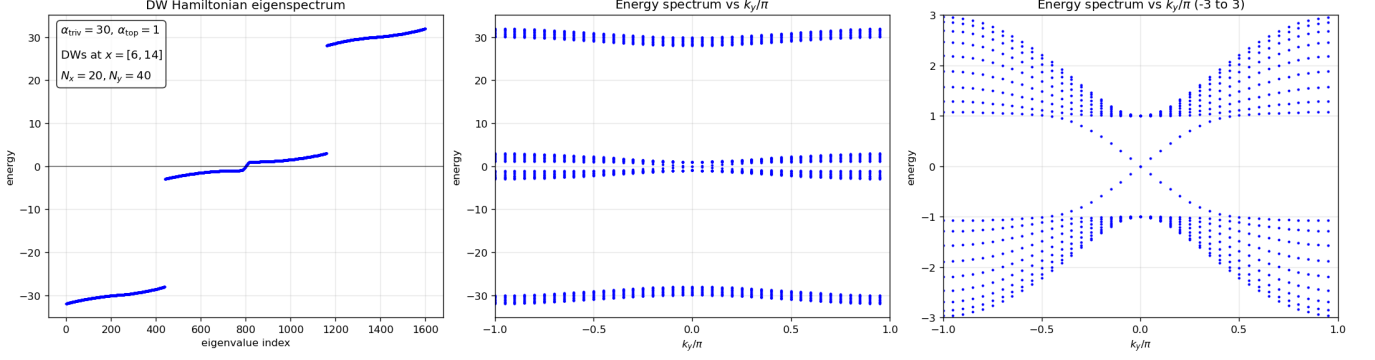


FIG. 1. Exact periodic-boundary spectrum for the benchmark geometry $N_x = 20$, $N_y = 40$, with $\alpha_{\text{triv}} = 30$, $\alpha_{\text{top}} = 1$, and domain walls at $x \in [6, 14]$. Left: ordered eigenvalues of the real-space Hamiltonian. Middle: spectrum of the k_y -resolved blocks obtained from the transform (16). Right: the same spectrum restricted to a low-energy window.

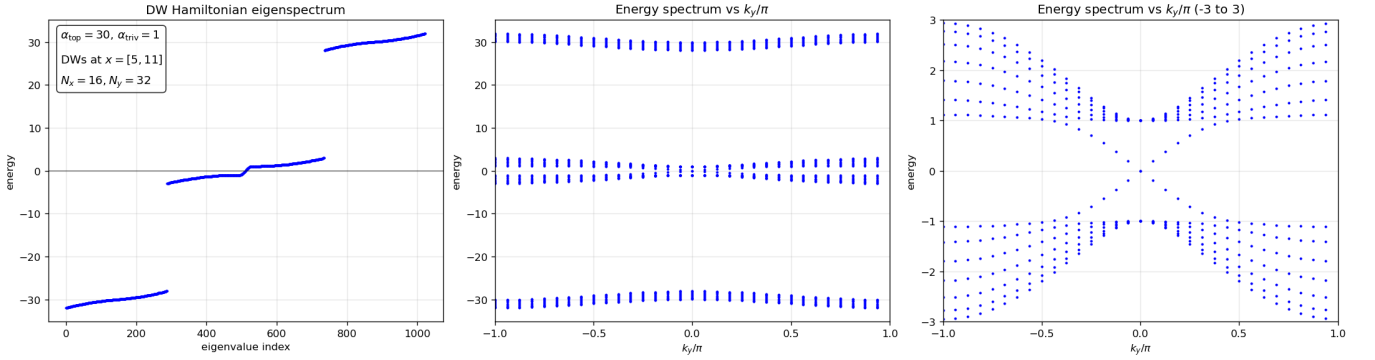


FIG. 2. The same three-panel spectral diagnostic as in Fig. 1, but for an alternate benchmark geometry $N_x = 16$, $N_y = 32$, with domain walls at $x \in [5, 11]$. This archived figure records an earlier system-size choice used during the development of the notebook.

IV. REDUCED COVARIANCE, ENTROPY, AND ENTANGLEMENT CONTOUR

Given a subsystem A , the notebook forms the reduced covariance by selecting the corresponding site-orbital indices,

$$G_A = G[\text{idx}(A), \text{idx}(A)]. \quad (17)$$

The associated one-body correlation matrix is

$$C_A = \frac{1}{2}(\mathbb{I}_A + G_A). \quad (18)$$

If $\{\lambda_n\}$ denotes the spectrum of C_A , the von Neumann entropy is

$$S(A) = -\sum_n \left[\lambda_n \log \lambda_n + (1 - \lambda_n) \log(1 - \lambda_n) \right]. \quad (19)$$

The notebook's entanglement contour is the diagonal of the operator

$$F_A = -C_A \log C_A - (\mathbb{I}_A - C_A) \log(\mathbb{I}_A - C_A), \quad (20)$$

namely

$$s_A(i) = (F_A)_{ii}. \quad (21)$$

Using the two-orbital structure, the contour displayed on the lattice is

$$s_A(x, y) = \sum_{\mu=0}^1 s_A(\mu, x, y), \quad (22)$$

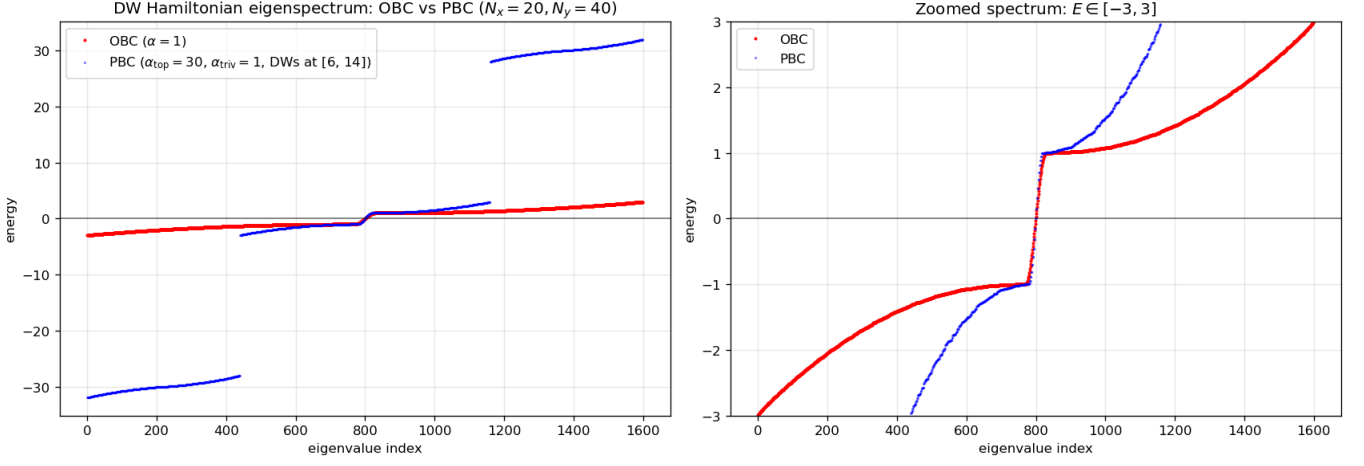


FIG. 3. Comparison between the periodic domain-wall benchmark spectrum (blue) and an open-boundary reference spectrum (red) used in the notebook. The left panel shows the full ordered spectrum; the right panel zooms into the window $E \in [-3, 3]$.

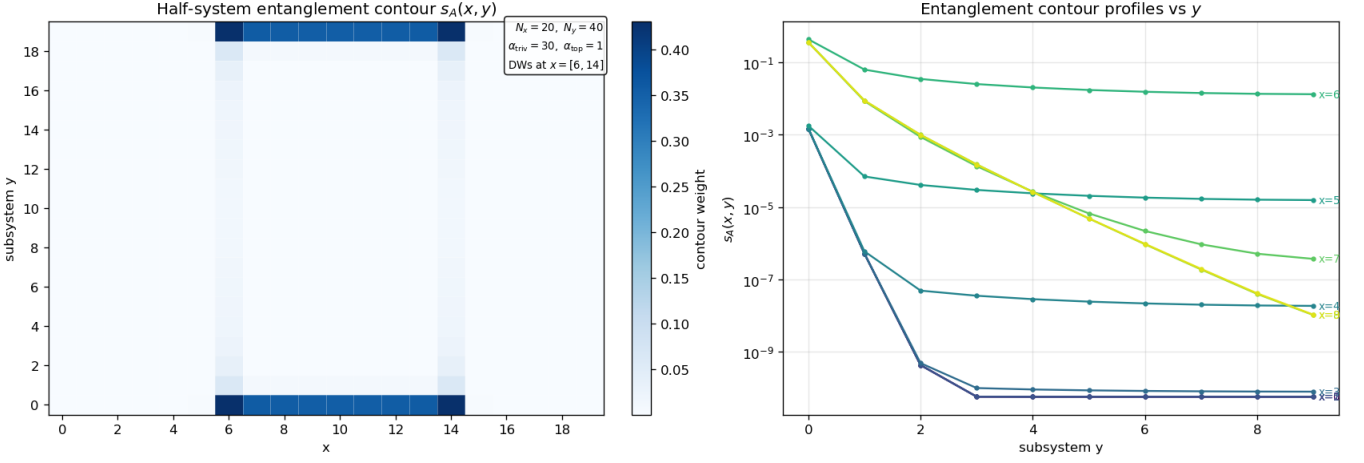


FIG. 4. Entanglement-contour diagnostics for the half-system strip. Left: the contour map $s_A(x, y)$ on a full-width subsystem of height $N_y/2$. Right: semilogarithmic profiles $s_A(x, y)$ versus the subsystem coordinate y for several fixed values of x .

which obeys the sum rule

$$\sum_{(x,y) \in A} s_A(x, y) = S(A). \quad (23)$$

For the half-system benchmark one chooses

$$A_{1/2} = \{(x, y) \mid 0 \leq x < N_x, y \in Y_{1/2}\}, \quad (24)$$

with $Y_{1/2}$ a contiguous block of length $N_y/2$ generated by the helper `contiguous_y_block`. The notebook then plots both the full map $s_{A_{1/2}}(x, y)$ and semilogarithmic cuts at fixed x .

V. FIXED- x SQUARED-CORRELATION PROFILES

The notebook also studies a boundary-sensitive equal-time two-point diagnostic constructed from the exact covariance G_{eq} at fixed x . For a given slice x_0 and y separation r_y , it forms the y -averaged squared correlator

$$C_G(x_0, r_y) = \frac{1}{N_y} \sum_{y=0}^{N_y-1} \sum_{\mu, \nu=0}^1 |G_{\text{eq}}(x_0, y, \mu; x_0, y + r_y, \nu)|^2, \quad (25)$$

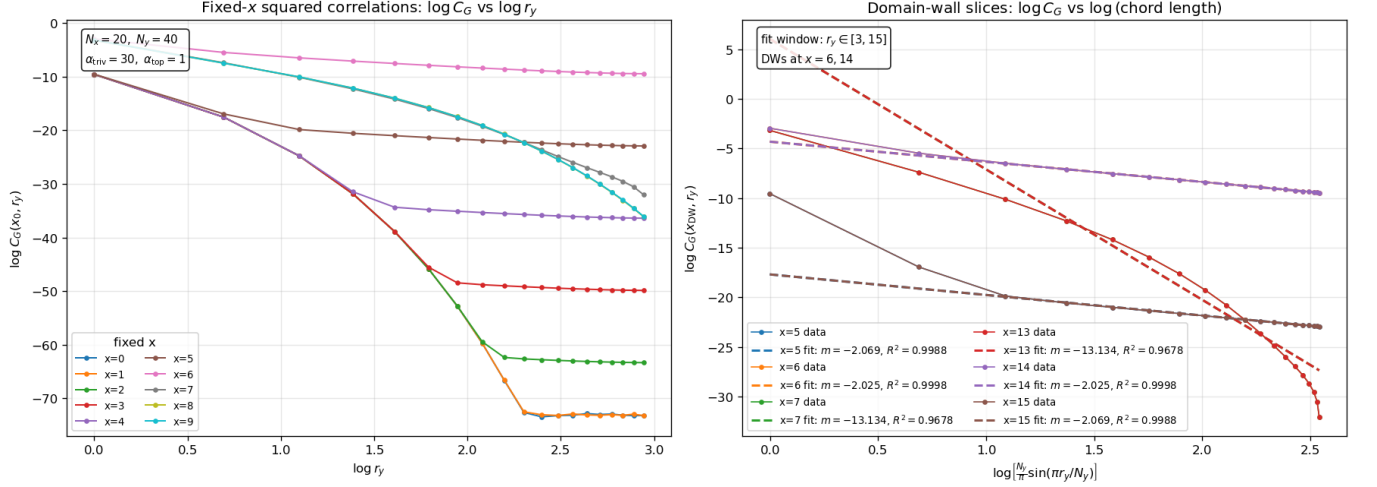


FIG. 5. Fixed- x squared-correlation benchmark built from the exact covariance G_{eq} . Left: $\log C_G(x_0, r_y)$ versus $\log r_y$ for several fixed- x slices, showing that the decay profile depends strongly on distance from the domain walls. Right: the domain-wall slices and their nearest neighbors plotted against the logarithm of the periodic chord length (26), together with linear fits over the window $r_y \in [3, 15]$. The domain-wall slices $x = 6, 14$ and their outer neighbors $x = 5, 15$ are close to a $C_G \sim \ell_y^{-2}$ law, while the inner neighbors $x = 7, 13$ decay much more rapidly.

with $y + r_y$ understood modulo N_y . The left panel of the updated benchmark figure presents $\log C_G(x_0, r_y)$ versus $\log r_y$ for a representative set of fixed- x slices. This broad survey is not intended as a precision fit; it is a quick way to see which x positions behave edge-like and which do not.

To test chord-length scaling more sharply, the notebook then isolates the domain-wall slices and their immediate neighbors and re-expresses the horizontal axis in terms of the periodic chord variable

$$\log \ell_y(r_y), \quad \ell_y(r_y) = \frac{N_y}{\pi} \sin\left(\frac{\pi r_y}{N_y}\right). \quad (26)$$

Fits are performed only on the intermediate window $r_y \in [3, 15]$, where the finite-size endpoint effects are visibly reduced. The figure shows that the actual domain-wall slices $x = 6$ and $x = 14$ are extremely well described by a straight line in $\log C_G$ versus $\log \ell_y$, with fitted slope $m \approx -2.025$ and $R^2 \approx 0.9998$. The exterior neighbors $x = 5, 15$ behave similarly, with slope $m \approx -2.069$ and $R^2 \approx 0.9988$. By contrast, the interior neighbors $x = 7, 13$ fall much more steeply, with fit slope $m \approx -13.134$ and noticeably poorer linearity. The practical conclusion is therefore that the clean algebraic chord-length scaling is localized on the domain wall and its immediate exterior neighbor, whereas slices shifted inward toward the topological strip are already dominated by much faster decay.

To test how much of that edge scaling survives after spatial averaging, the notebook also forms the uniform x average of the same fixed-slice correlator,

$$C_{\text{avg}}(r_y) = \frac{1}{N_x} \sum_{x=0}^{N_x-1} C_G(x, r_y), \quad (27)$$

and analyzes it on both $\log r_y$ and log-chord axes. The resulting x -averaged profile is shown in Fig. 7. Even after including all x slices, the intermediate-range fit over $r_y \in [3, 15]$ remains very close to a straight line in $\log C_{\text{avg}}$ versus $\log \ell_y$, with slope $m \approx -2.069$ and $R^2 \approx 0.9990$. This is interesting because the strictly local fit diagnostics of Fig. 6 show that only a narrow set of slices near the two interfaces display clean edge-like scaling. The global average therefore suggests that those interface-dominated contributions still control the intermediate-distance behavior of the spatially averaged squared correlator, even though most individual off-wall slices decay much faster.

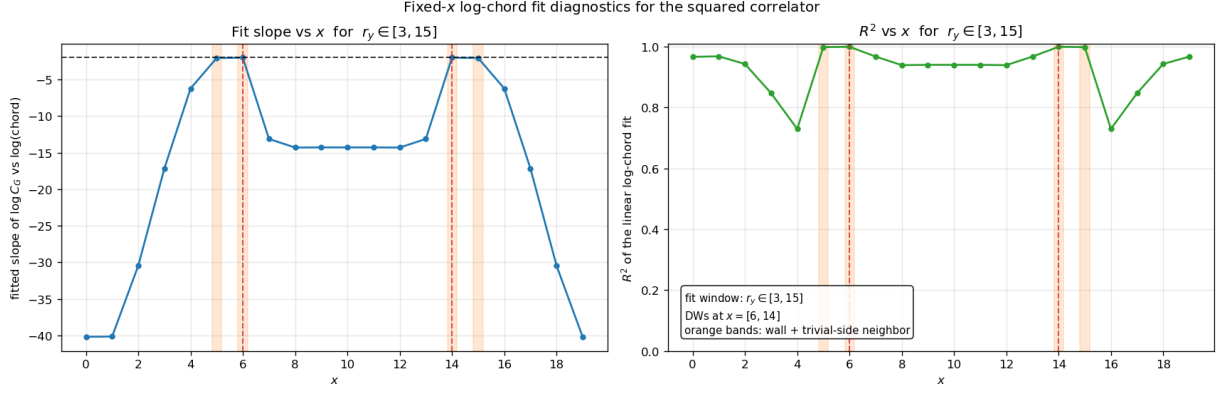


FIG. 6. Fixed- x log-chord fit diagnostics extracted from the fit window $r_y \in [3, 15]$. Left: fitted slope of $\log C_G(x, r_y)$ versus $\log \ell_y(r_y)$ as a function of x , with a dashed horizontal reference line at slope -2 and vertical markers at the two domain walls. Right: the corresponding coefficient of determination R^2 as a function of x . Together these panels show that the near- -2 chord-length scaling is sharply localized on the wall and the trivial-side neighbor: for the left interface the pair $(x = 5, 6)$ behaves similarly, while for the right interface the analogous pair is $(x = 14, 15)$.

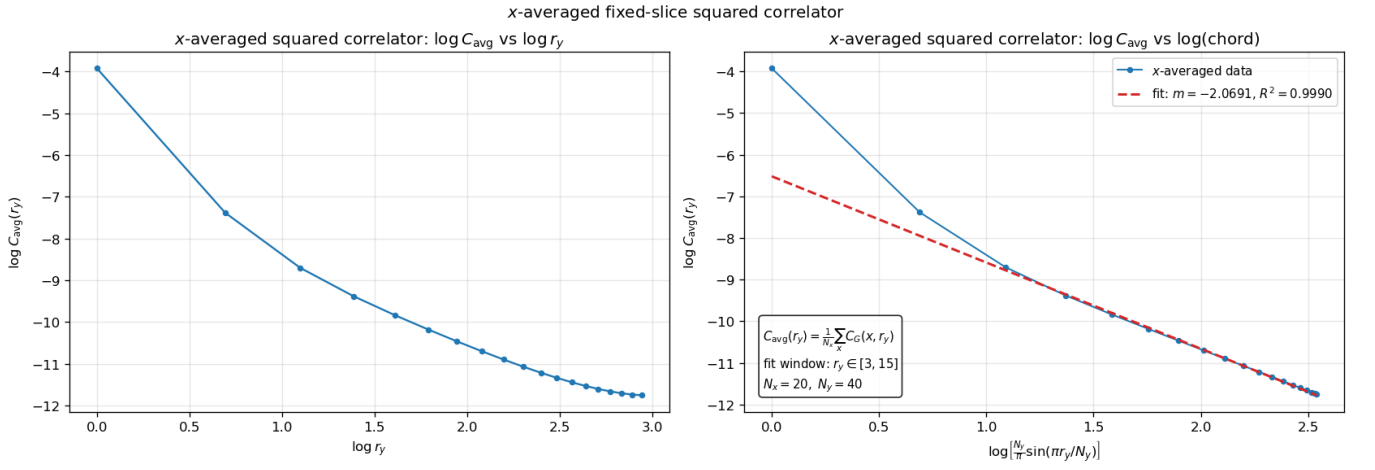


FIG. 7. Uniformly x -averaged squared-correlation diagnostic built from the fixed-slice correlator of Eq. (25). Left: $\log C_{\text{avg}}(r_y)$ versus $\log r_y$, where $C_{\text{avg}}(r_y) = N_x^{-1} \sum_x C_G(x, r_y)$. Right: the same averaged data plotted against the logarithm of the periodic chord length (26), together with the linear fit over the window $r_y \in [3, 15]$. The fitted slope remains close to -2 , indicating that the domain-wall contribution continues to dominate this averaged observable over the intermediate distances resolved in the benchmark.

VI. LATTICE AND MODULAR CURRENT DENSITIES

The current-density constructions used alongside the notebook can be organized starting from a general quadratic lattice Hamiltonian

$$H = \sum_{\mathbf{r}, \mathbf{r}'} \Psi_{\mathbf{r}}^{\dagger} h_{\mathbf{r}\mathbf{r}'} \Psi_{\mathbf{r}'}, \quad \mathbf{r} = (x, y), \quad (28)$$

where each block $h_{\mathbf{r}\mathbf{r}'}$ is a 2×2 matrix in the orbital indices $\mu, \nu \in \{0, 1\}$. The cell occupation operator is

$$n_{\mathbf{r}} = \Psi_{\mathbf{r}}^{\dagger} \Psi_{\mathbf{r}}. \quad (29)$$

Using the Heisenberg equation, one finds a lattice continuity equation of the form

$$\dot{n}_{\mathbf{r}} + \sum_{\mathbf{r}'} J_{\mathbf{r} \rightarrow \mathbf{r}'} = 0, \quad (30)$$

with bond-current operator

$$J_{\mathbf{r} \rightarrow \mathbf{r}'} = -i \Psi_{\mathbf{r}}^\dagger h_{\mathbf{r}\mathbf{r}'} \Psi_{\mathbf{r}'} + i \Psi_{\mathbf{r}'}^\dagger h_{\mathbf{r}'\mathbf{r}} \Psi_{\mathbf{r}}. \quad (31)$$

For a Gaussian state with one-body correlation matrix

$$C_{\mathbf{r}\mathbf{r}'}^{\mu\nu} = \langle c_{\mathbf{r},\mu}^\dagger c_{\mathbf{r}',\nu} \rangle, \quad (32)$$

Hermiticity $h_{\mathbf{r}'\mathbf{r}} = h_{\mathbf{r}\mathbf{r}'}^\dagger$ reduces the expectation value of Eq. (31) to

$$j_{\mathbf{r} \rightarrow \mathbf{r}'} = 2 \operatorname{Im} \operatorname{Tr} [h_{\mathbf{r}\mathbf{r}'} C_{\mathbf{r}'\mathbf{r}}]. \quad (33)$$

Written explicitly in the two-orbital basis,

$$j_{\mathbf{r} \rightarrow \mathbf{r}'} = 2 \operatorname{Im} \sum_{\mu,\nu=0}^1 h_{\mathbf{r}\mathbf{r}'}^{\mu\nu} C_{\mathbf{r}'\mathbf{r}}^{\nu\mu}. \quad (34)$$

On the square lattice of Eqs. (6) and (7), the positively oriented nearest-neighbor currents are therefore

$$j_x(x, y) = 2 \operatorname{Im} \operatorname{Tr} [h_{(x,y),(x+1,y)} C_{(x+1,y),(x,y)}], \quad (35)$$

$$j_y(x, y) = 2 \operatorname{Im} \operatorname{Tr} [h_{(x,y),(x,y+1)} C_{(x,y+1),(x,y)}]. \quad (36)$$

For the physical domain-wall Hamiltonian one simply substitutes the fixed nearest-neighbor blocks $h_{(x,y),(x+1,y)} = t_x$ and $h_{(x,y),(x,y+1)} = t_y$ from Eq. (7). In the code this is the gauge-fixed lattice-current convention associated with the flattening map $i(\mu, x, y) = \mu + 2x + 2N_x y$ and the orbital basis defined in Eq. (8).

The same formula applies to modular currents on a subsystem A . From the reduced covariance Eq. (17) and correlation matrix Eq. (18), the Gaussian reduced density matrix can be written as

$$\rho_A \propto \exp[-\Psi_A^\dagger h_A^{\text{mod}} \Psi_A], \quad (37)$$

where the one-body modular Hamiltonian is

$$h_A^{\text{mod}} = \log[(\mathbb{I}_A - C_A) C_A^{-1}]. \quad (38)$$

Using $C_A = (\mathbb{I}_A + G_A)/2$, this is equivalently

$$h_A^{\text{mod}} = \log[(\mathbb{I}_A - G_A)(\mathbb{I}_A + G_A)^{-1}] = -2 \operatorname{arctanh}(G_A), \quad (39)$$

which is the form used by the auxiliary modular-current script. The modular current density is then obtained by replacing $h_{\mathbf{r}\mathbf{r}'}$ in Eqs. (35)–(36) by the corresponding blocks of h_A^{mod} , while keeping the same reduced correlator C_A . Thus the physical and modular current constructions differ only in the one-body generator inserted into the same continuity-equation formula.

VII. LOCAL CHERN MARKER

Topological structure is probed through the local Chern marker built from the projector

$$P = \frac{1}{2}(\mathbb{I} + G). \quad (40)$$

Writing X and Y for the position operators, the notebook evaluates the six-index operator

$$M = 2\pi i (PXPYP - PYPXP), \quad (41)$$

and then defines the site-resolved marker

$$C(x, y) = \sum_{\mu=0}^1 M_{(x,y,\mu),(x,y,\mu)}. \quad (42)$$

The stored plot uses the bounded visualization

$$C_{\tanh}(x, y) = \tanh(C(x, y)). \quad (43)$$

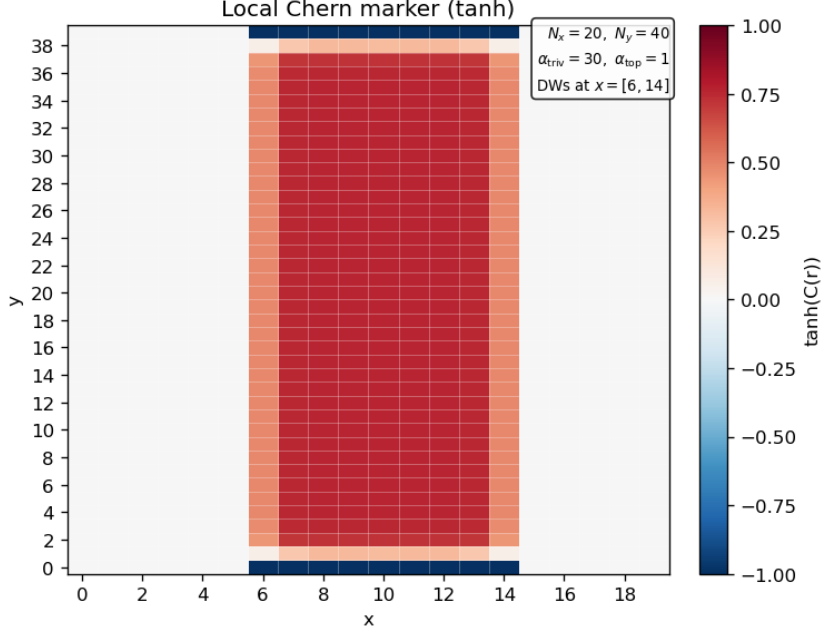


FIG. 8. Tanh-compressed local Chern marker $C_{\tanh}(x, y)$ for the exact Gaussian ground state. The nearly quantized domain-wall slab is visibly localized between the two marked interfaces.

VIII. INTEGRATED ENTANGLEMENT CONTOUR AND CHORD-LENGTH FITS

For the contour benchmark as a function of subsystem height, the notebook fixes the full-width strip

$$A(A_y) = \{(x, y) \mid 0 \leq x < N_x, 0 \leq y < A_y\}, \quad 1 \leq A_y \leq N_y - 1, \quad (44)$$

computes the corresponding contour $s_{A(A_y)}(x, y)$, and only *afterward* restricts the final sum to a chosen x window I_x . The integrated quantity is therefore

$$S_{I_x}(A_y) = \sum_{x \in I_x} \sum_{y=0}^{A_y-1} s_{A(A_y)}(x, y). \quad (45)$$

This distinction is crucial: the subsystem itself remains full width in x , while the interval I_x serves only as a diagnostic window for where the entanglement contour is spatially concentrated.

To test the expected one-dimensional edge scaling, the notebook separates Eq. (45) into the two branches

$$A_y < \frac{N_y}{2}, \quad A_y > \frac{N_y}{2}, \quad (46)$$

because the map

$$A_y \mapsto \log\left[\sin\left(\frac{\pi A_y}{N_y}\right)\right] \quad (47)$$

is not one-to-one on the full interval $1 \leq A_y \leq N_y - 1$. Each branch is then fit to

$$S_{I_x}(A_y) \simeq m_{I_x} \log\left[\sin\left(\frac{\pi A_y}{N_y}\right)\right] + b_{I_x}. \quad (48)$$

IX. FLATTENED-HAMILTONIAN TRIAL-ORBITAL BENCHMARK

The notebook also benchmarks an overcomplete-Wannier surrogate for the exact ground-state covariance by first constructing a flattened single-particle Hamiltonian from the trial-orbital projectors generated by `construct_OW_projectors`.

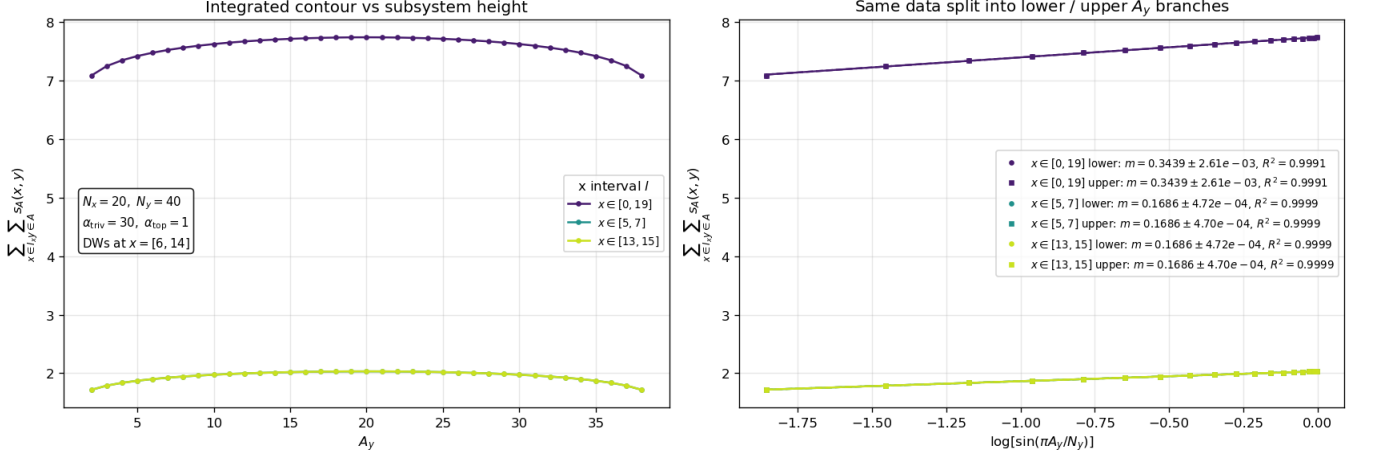


FIG. 9. Integrated contour benchmark. Left: $S_{I_x}(A_y)$ for the full-width sum and for two narrow domain-wall-centered windows. Right: the same data split into lower and upper A_y branches and fit to the log-chord form (48).

For a choice of trial spinor $\tau \in \{\sigma_X, \sigma_Y, \sigma_Z\}$, the notebook forms

$$H_b^{(\tau)} = \sum_R \left[P_{A,R}^{+(\tau)} + P_{B,R}^{+(\tau)} - P_{A,R}^{-(\tau)} - P_{B,R}^{-(\tau)} \right], \quad (49)$$

followed by an explicit Hermitization,

$$H_b^{(\tau)} \longrightarrow \frac{1}{2} \left(H_b^{(\tau)} + H_b^{(\tau)\dagger} \right). \quad (50)$$

After diagonalizing

$$H_b^{(\tau)} v_n^{(\tau)} = \varepsilon_n^{(\tau)} v_n^{(\tau)}, \quad (51)$$

it defines the occupied projector

$$P_{\text{occ}}^{(\tau)} = \sum_{\varepsilon_n^{(\tau)} < 0} \left| v_n^{(\tau)} \right\rangle \left\langle v_n^{(\tau)} \right|, \quad (52)$$

and then uses the same flattened covariance convention as in the exact benchmark,

$$G_b^{(\tau)} = 2P_{\text{occ}}^{(\tau)} - \mathbb{I}. \quad (53)$$

These trial-orbital covariances are then passed through exactly the same entanglement-contour pipeline as the exact G_{eq} .

The motivation for this surrogate is dynamical as well as variational. If the post-selected circuit and the adaptive circuit relax to the same steady state, then the relaxation time of the adaptive dynamics is bounded from below by the post-selected relaxation rate. In that case, understanding the slow approach to the steady state can be reduced, at least at the level of a lower bound, to understanding the effective post-selected generator encoded by the flattened Hamiltonian. The notebook therefore uses $H_b^{(\tau)}$ as a controlled proxy for the long-time structure that may govern slow purification.

A second motivation is that, because the construction is based on overcomplete-Wannier orbitals rather than exact band projectors, the resulting spectrum is not perfectly flat. Instead it is modulated by the L^2 -normalized form factor of the chosen Wannier envelope. This makes the trial-orbital dependence physically informative rather than merely cosmetic: different choices of τ probe how sensitive the surrogate edge theory is to the spatial and orbital structure built into the OW ansatz.

Figure 10 summarizes the single-particle spectra of $H_b^{(\tau)}$. The $\tau \in \sigma_X$ and $\tau \in \sigma_Y$ choices are nearly indistinguishable: both show a strongly flattened ordered spectrum with most eigenvalues clustered near ± 2 , together with a pair of dispersing k_y -resolved branches crossing near zero. By contrast, $\tau \in \sigma_Z$ produces a much broader spectrum,

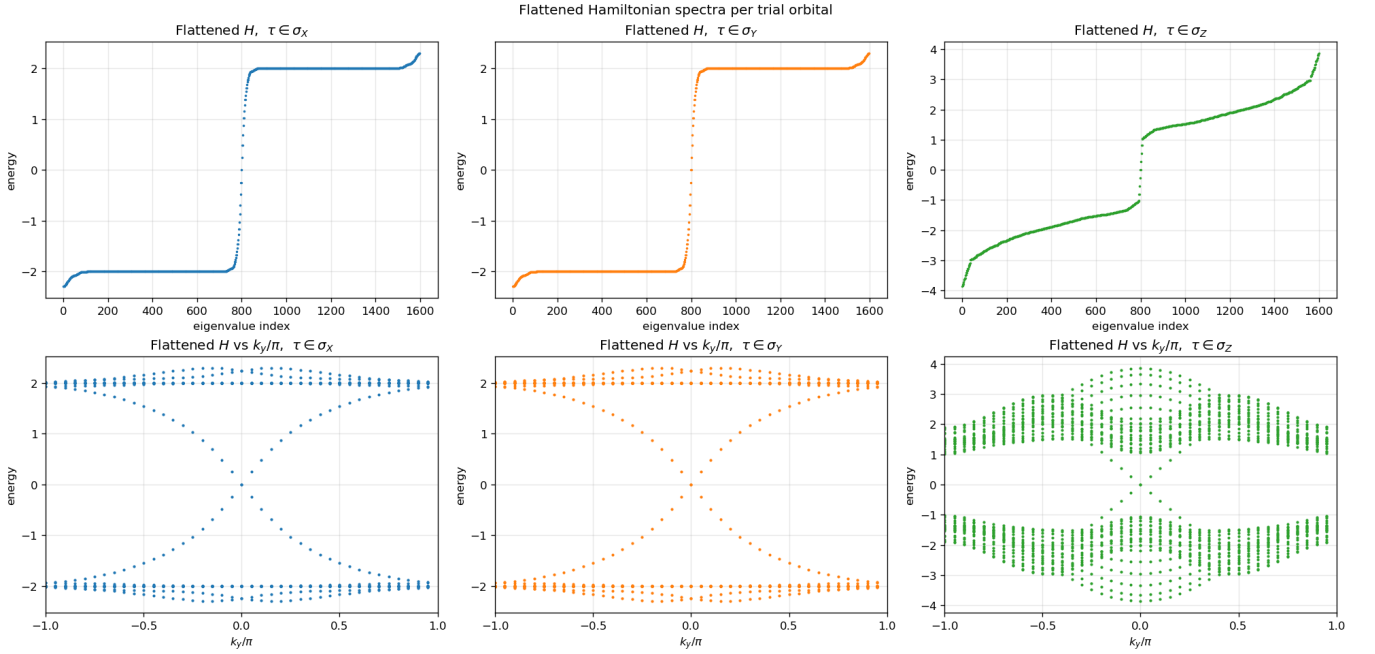


FIG. 10. Spectra of the OW flattened Hamiltonian $H_b^{(\tau)}$ for the three trial-orbital choices $\tau \in \sigma_X, \sigma_Y, \sigma_Z$. Top row: ordered eigenvalues of the Hermitized matrix (50). Bottom row: spectra of the corresponding k_y -resolved blocks, obtained by the same Fourier-block decomposition used for the exact benchmark. The X and Y trial orbitals lead to nearly identical flattened spectra, whereas the Z choice is appreciably broader and more dispersive.

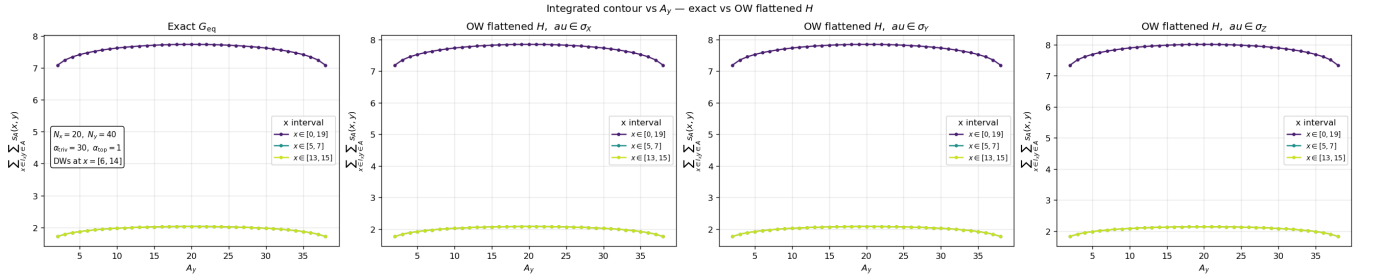


FIG. 11. Integrated contour curves $S_{I_x}(A_y) = \sum_{x \in I_x} \sum_{y=0}^{A_y-1} s_{A(A_y)}(x, y)$ for the exact covariance and for the three trial-orbital flattened-H covariances. The four panels compare the full-width diagnostic window and the two domain-wall-centered windows used in the exact benchmark. Across all three choices of τ , the raw contour profiles remain close to the exact curves.

extending to roughly ± 4 , and a visibly different k_y -resolved band structure. In that sense the X and Y trial spinors generate very similar OW surrogates, whereas the Z choice is spectrally distinct already at the level of the flattened Hamiltonian.

The entanglement benchmark based on these trial-orbital covariances is shown in Figs. 11 and 12. For each $G_b^{(\tau)}$, the notebook computes the full-width contour on the strip $A(A_y) = [0, N_x] \times [0, A_y]$ and then integrates the resulting contour over the same three x windows used in the exact benchmark: the full width $I_x = [0, N_x - 1]$ and the two three-site windows centered on the domain walls. The raw A_y -dependence in Fig. 11 already shows that all three trial-orbital constructions track the exact benchmark closely.

The more discriminating comparison comes from the branch-by-branch log-chord fits. Figure 12 shows that the X and Y trial orbitals reproduce the exact fitted slopes extremely well. For the full-width interval, the exact benchmark gives $m \approx 0.3439$, while the X and Y trial orbitals give $m \approx 0.3448$. For the two domain-wall-centered windows, the exact slope is $m \approx 0.1686$ on both branches, compared with $m \approx 0.1691$ for X and Y . The $\tau \in \sigma_Z$ construction still shows the same qualitative additivity of two nearly equal wall contributions, but its slopes are more noticeably shifted, to about 0.3511 on the full-width window and 0.166–0.167 on the narrow wall windows. Thus the integrated entanglement-contour scaling is quite robust under the OW approximation, with X and Y performing especially well

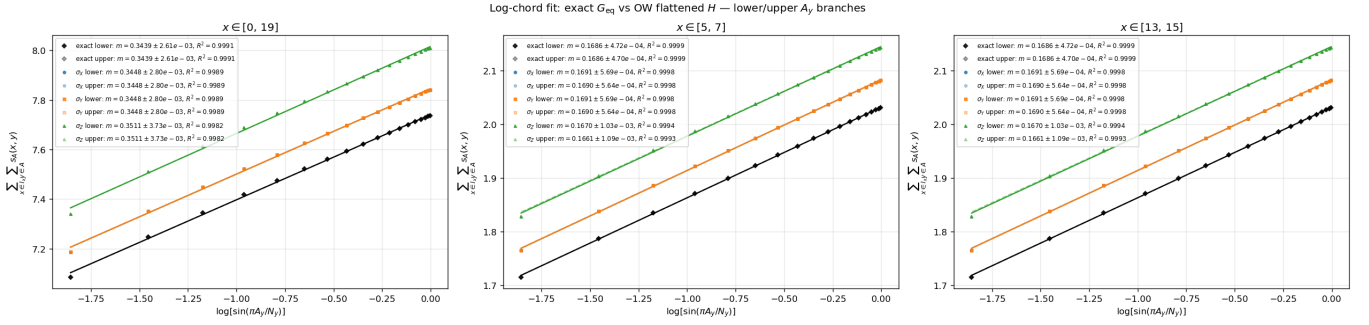


FIG. 12. Lower- and upper-branch log-chord fits of the integrated contour for the exact covariance and the three OW flattened-H trial-orbital constructions. Each panel corresponds to one x interval: full width, left domain wall, and right domain wall. The X and Y trial orbitals reproduce the exact fit slopes almost indistinguishably, while the Z trial orbital shows a larger but still controlled deviation.

and Z remaining qualitatively correct but quantitatively less accurate.

X. CONTOUR-RESTRICTED MUTUAL-INFORMATION DIAGNOSTICS

The notebook also implements a multipartite diagnostic adapted from `analysis.ipynb`. One chooses three contiguous intervals in the y direction,

$$A, \quad B, \quad C, \quad (54)$$

of lengths L_A , L_B , and L_C , respectively, and either fixes their anchor y_0 or averages over all y_0 translations around the y circle. The subsystems remain full width in x ; the chosen interval I_x is applied only in the final contour sum.

For any y region Y , let $s_Y(x, y)$ denote the contour computed on the full-width subsystem associated with Y . The notebook then defines the contour-restricted entropy proxy

$$S^{(I_x)}(Y) = \sum_{x \in I_x} \sum_{y \in Y} s_Y(x, y). \quad (55)$$

From these quantities it constructs

$$I_2^{(I_x)}(A : C) = S^{(I_x)}(A) + S^{(I_x)}(C) - S^{(I_x)}(AC), \quad (56)$$

$$I_3^{(I_x)}(A : B : C) = S^{(I_x)}(A) + S^{(I_x)}(B) + S^{(I_x)}(C) - S^{(I_x)}(AB) - S^{(I_x)}(AC) - S^{(I_x)}(BC) + S^{(I_x)}(ABC). \quad (57)$$

The benchmark emphasizes the difference

$$\Delta I^{(I_x)} \equiv I_2^{(I_x)} - I_3^{(I_x)}. \quad (58)$$

To compare with a one-dimensional conformal variable, the notebook uses the chord length

$$\ell(L) = \frac{N_y}{\pi} \sin\left(\frac{\pi L}{N_y}\right), \quad (59)$$

and the cross ratio

$$x = \frac{\ell(L_A) \ell(L_C)}{\ell(L_A + L_B) \ell(L_B + L_C)}. \quad (60)$$

The final linear fit is performed against

$$\Delta I^{(I_x)}(x) \simeq m_{I_x} \log\left(\frac{1}{1-x}\right) + b_{I_x}. \quad (61)$$

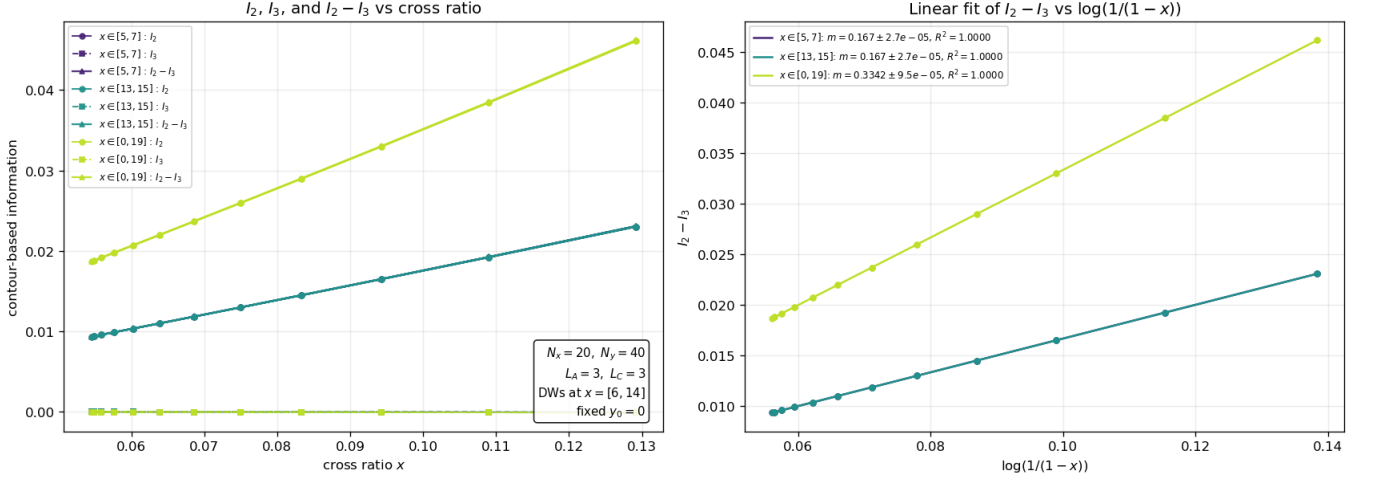


FIG. 13. Contour-restricted multipartite-information benchmark for three different x windows. Left: $I_2^{(I_x)}$, $I_3^{(I_x)}$, and $\Delta I^{(I_x)} = I_2^{(I_x)} - I_3^{(I_x)}$ plotted against the cross ratio (60). Right: linear fits of $\Delta I^{(I_x)}$ against $\log(1/(1-x))$.

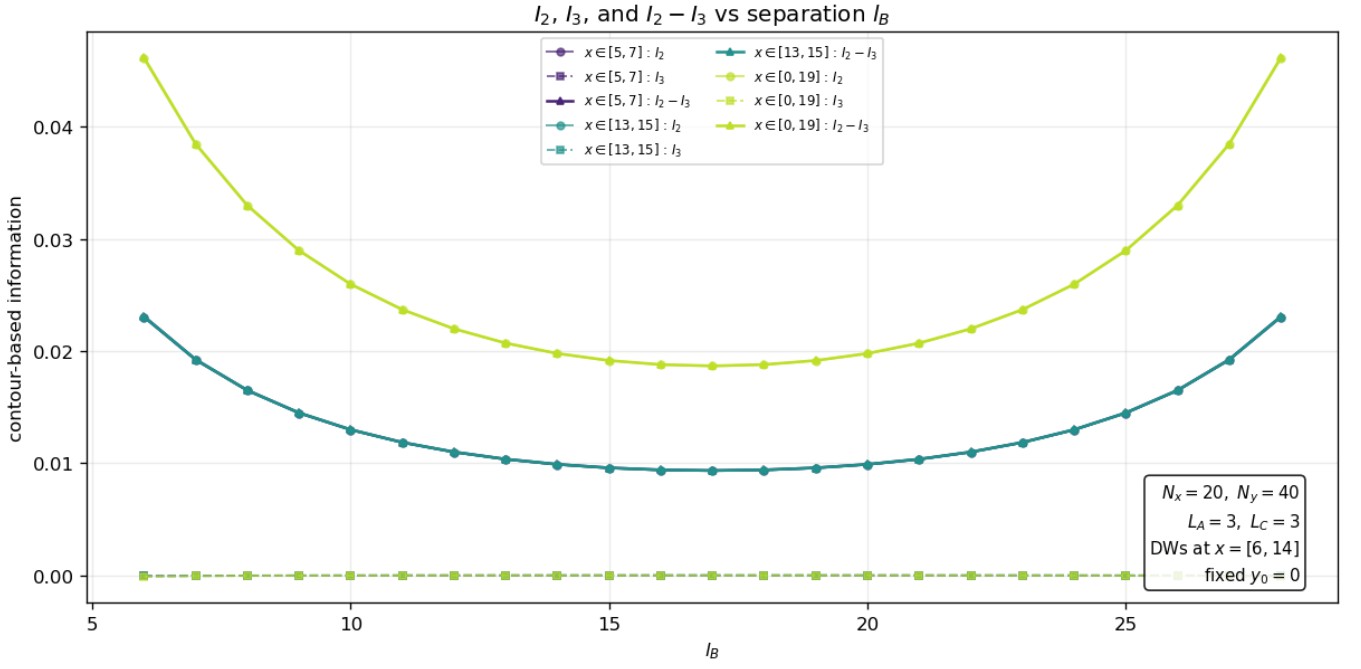


FIG. 14. The same contour-restricted information measures displayed directly as functions of the separation l_B between the two outer intervals of lengths L_A and L_C . The figure illustrates how the signal depends on the choice of the post-contour x window.

XI. ANCILLARY BENCHMARK METRICS

Although the presently saved figure set does not include the dynamic-state comparison cell, the notebook helper layer also defines two standard diagnostics for comparing a dynamically prepared covariance G_{dyn} against the equilibrium benchmark G_{eq} .

First, the Frobenius mismatch is

$$\|G_{\text{eq}} - G_{\text{dyn}}\|_F = \sqrt{\text{Tr}[(G_{\text{eq}} - G_{\text{dyn}})^\dagger (G_{\text{eq}} - G_{\text{dyn}})]}. \quad (62)$$

Second, the notebook defines a site-resolved difference map by reshaping the matrix difference into a six-index tensor

and tracing its squared modulus over all source and orbital indices,

$$d(x, y) = \left[\sum_{\mu, \nu, x', y'} |\Delta G_{\mu, x, y; \nu, x', y'}|^2 \right]^{1/2}, \quad \Delta G = G_{\text{eq}} - G_{\text{dyn}}. \quad (63)$$

These metrics are natural Gaussian benchmarks should the dynamic-preparation portion of the notebook be reactivated.

XII. CLOSING REMARKS

The benchmark notebook is conceptually simple but diagnostically rich. Everything begins from the exact Gaussian ground state of the real-space Hamiltonian (6), yet the subsequent observables resolve the same state in complementary ways: by ordered single-particle energies, by k_y -resolved spectral blocks, by fixed- x squared-correlation profiles, by the local entropy density encoded in the entanglement contour, by the local Chern response, by strip-integrated log-chord scaling, and by contour-restricted multipartite information.

For practical use, the most important structural point is the following. Whenever the notebook loops over an x interval, the intended construction is usually *not* to redefine the subsystem in x ; rather, one computes the entanglement contour on a subsystem that is full width in x and only then restricts the final sum to a post-processed window I_x . This is precisely the convention encoded in Eqs. (45) and (55), and it is the convention that gives the cleanest physical interpretation of the benchmark figures.